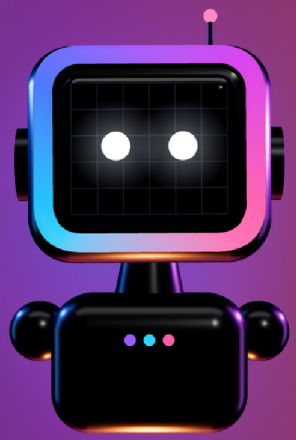
TECHNICAL REQUIREMENTS DOCUMENT · MVP · META GLOBAL AI DEVELOPER HACKATHON 2026

RoboCraft Lab

Describe a machine by text or voice. **Meta Muse Spark** engineers it from **real, buyable parts**, wires it, and writes its logic; a physics engine **checks every claim**; you watch it run in 3D before buying a single part.

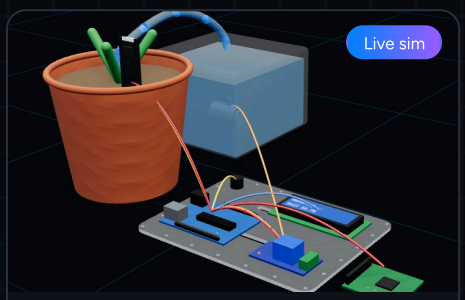


Powered by Meta Muse · Track · Push the Frontier · Muse Spark 1.3 · SAM 3.1 · Muse Image 1.0 · C++17 → WebAssembly

TEAM: MetaRobo

MEMBERS: Samuel Aboderin

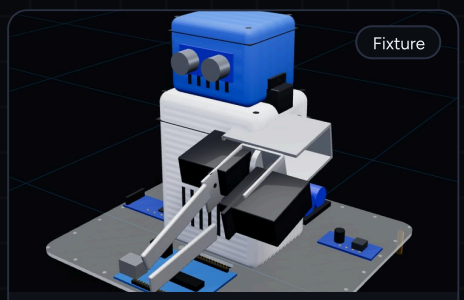
REPOSITORY: github.com/aboderinsamuel/MetaRobo



Smart Plant Waterer — Automation

Soil probe → controller rule → relay → pump. Water flows down the tube, soil darkens, LCD shows "Watering…".

10 parts · 10 wires · 0.4 A peak · wiring ✓ logic ✓



Waving Desk Robot — Mechanism

Limbs built like a real servo kit: servo → horn → bracket → servo. Torque, speed and current simulated per actuator.

16 parts · 2 joints · MG996R · PCA9685

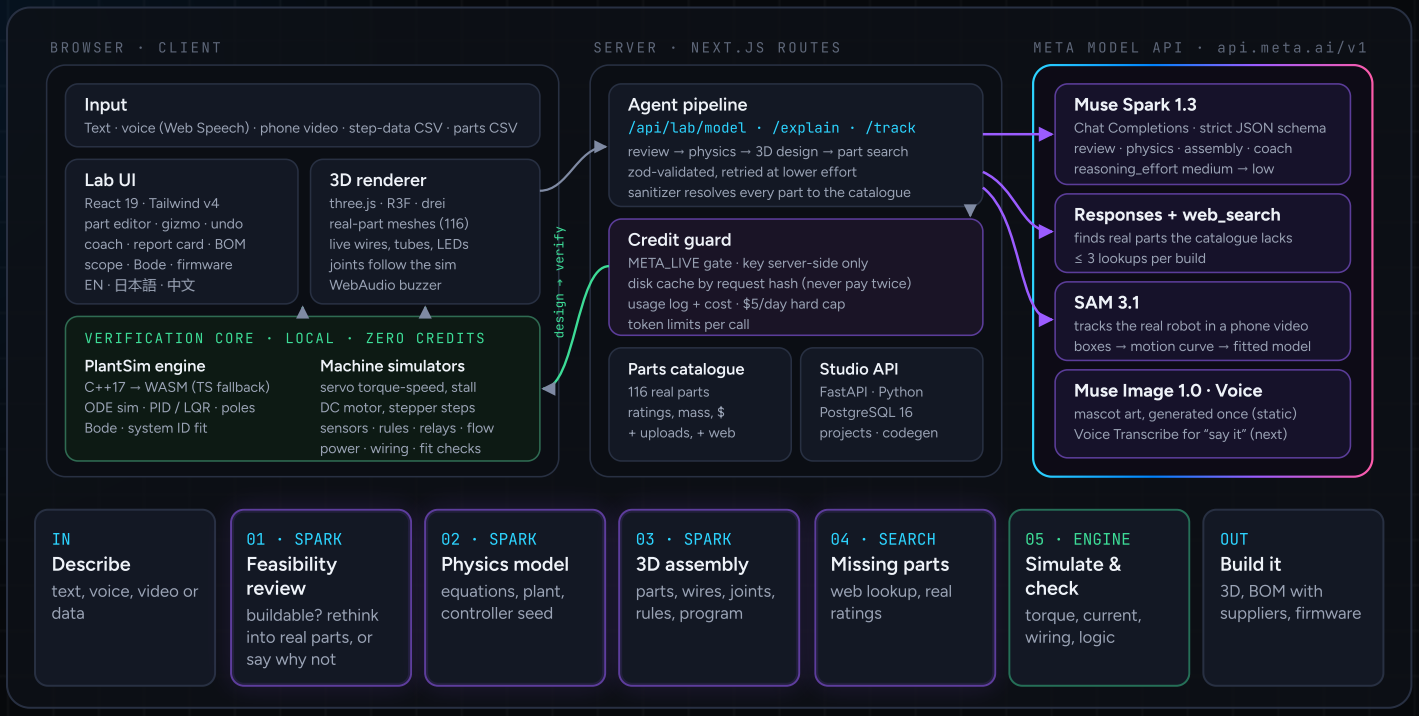


Robo, in Every Style — Muse Image

The mascot in 8 art styles, generated once and served statically. EN · 日本語 · 中文 interface with spoken intro.

8 images · $0.08 total · 3 languages

01 High-Level Architecture · one Next.js app, a WASM physics core, Meta models behind a credit-safe server gateway



The model proposes; the engine disposes. Muse Spark never touches the 3D scene directly: its JSON is **validated against the parts catalogue** (sizes, masses, prices and ratings come from real datasheets, never the model), then simulated. Every claim (strong enough, fast enough, powered and wired right) is a numeric check shown to the user, and the coach explains failures in plain words.

4
Meta models wired in: Muse Spark, web search, SAM 3.1, Muse Image

116
real catalogue parts with datasheet ratings, plus uploads and web lookups

≈ 15¢
per full AI build (review + physics + design + searches), cached after

112
automated tests; none ever call a paid API

02 Tools & Resources

TOOL / SERVICE	PURPOSE
Meta Model API `muse-spark-1.3`	Engineering review, physics model, full 3D assembly with wiring and controller logic, coaching. Strict JSON schemas.
Responses API `web_search tool`	Finds real, buyable parts the catalogue lacks and returns ratings, size, mass and supplier links.
SAM 3.1 `sam-3.1`	Tracks the real robot in a phone clip; the track becomes a measured response the engine fits a model to.
Muse Image `muse-image-1.0`	Generated the mascot in 8 styles once, offline ($0.08); served statically.
Emscripten	Compiles the C++17 PlantSim engine to WebAssembly for in-browser simulation.
Supplier links	Adafruit, SparkFun, Pololu, RobotShop, DigiKey, LCSC, Seeed: an orderable BOM.
Web Speech API	Browser dictation and the spoken intro (EN / JA / ZH voices).
Vitest · ESLint · GitHub	Tests on recorded fixtures, lint gates, dated commits; built with Claude Code.
Vercel · Docker	Public deploy (Next.js + FastAPI services); docker-compose for local Postgres.

SECURITY & PRIVACY

API key lives only in server env, never in the browser bundle or the repo. Uploaded part lists are schema-checked and size-capped; videos are deleted from Meta after tracking. No accounts and no personal data in the MVP; preferences stay in the visitor's own browser.

03 Technology Stack

FRONTEND

Next.js 16 · React 19 · TypeScript

App Router with Turbopack; one codebase for landing, Lab and server routes. Tailwind v4 + motion for a fast, animated UI.

3D

three.js · React Three Fiber · drei

Declarative scene graph that mirrors the assembly tree, so the AI's parent/child design renders and animates directly.

SIMULATION

C++17 → WASM · TypeScript sims

Native-speed ODE and control maths in the browser: instant re-simulation on every edit, no server cost.

BACKEND

Next.js route handlers · FastAPI

AI pipeline lives beside the UI (shared zod types); FastAPI keeps the Studio's Python engines and codegen.

DATA

Typed catalogue · PostgreSQL 16

Parts are versioned in code for exact ratings; Postgres stores Studio projects; disk cache + JSONL usage log for AI calls.

CONTRACTS

zod 4 → JSON Schema

One schema drives Muse Spark's strict output, runtime validation and TypeScript types; malformed output can't reach the 3D.

Why this shape: simulation must be instant and free, so it runs client-side in WASM; the AI must be safe and cheap, so it sits behind one server gateway with caching and a spend cap; the 3D must be trustworthy, so it renders only validated, catalogue-backed parts.

04 Assumptions, Dependencies & Constraints

- **Meta Model API availability.** Muse Spark, SAM 3.1 and web search reachable at api.meta.ai/v1; without a key or with `META_LIVE` off, the app says so, and demos still run offline.
- **$150 credit budget.** Live calls off by default, cached by request hash, logged with cost, capped at $5/day; a full build costs ≈ 10–20¢.
- **Latency.** A design call can take 30–90 s at medium reasoning; the UI streams pipeline progress and retries at lower effort on failure.
- **Browser.** WebGL 2 and WebAssembly required (any current Chrome, Edge, Safari, Firefox); speech input where the browser supports it.
- **Real parts only.** Designs must resolve to catalogue, fabricated (laser-cut, 3D-printed, extrusion, rod) or web-found parts; unresolved parts are flagged, never invented.
- **Simulation fidelity.** Rigid-body kinematics with datasheet actuator models and first-order behaviour (drift, flow, thresholds); not FEA or CFD. Results are engineering estimates.
- **Video input.** Clips ≤ 20 s with one moving part and a known length for scale; SAM results cached per video hash.
- **Branding & rules.** Plain-text "Powered by Meta Muse", no Meta logo; only work built during the hackathon is demoed.

05 MVP Scope · status at time of writing

● WORKING TODAY

- Text / voice → machine engineered from real parts
- Feasibility review: buildable variant or a clear "can't"
- 3D editor: pick, move, rotate, swap, undo
- Actuator, wiring and behaviour sims; report card; coach
- Priced BOM, supplier links, uploads, web lookup
- PID / LQR control lab with scope, Bode, firmware

◐ BUILDING NOW

- Change a built machine by text or voice ("add a Christmas tree and wrap the lights round it")
- Version timeline: rewind, fast-forward, branch
- LED patterns (blink, chase, twinkle, fade, rainbow) driving the 3D and generating matching Arduino, Verilog and VHDL

○ BEFORE SUBMISSION

- Muse Voice Transcribe as the voice path
- Live SAM 3.1 clip: measured vs simulated overlay
- Benchmark: first-try vs checked-and-fixed pass rate
- Public deploy, public repo, 3-minute demo video

